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Clinical and Incidental Perineural Invasion of Cutaneous Squamous Cell Carcinoma

A Systematic Review and Pooled Analysis of Outcomes Data

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IMPORTANCE Perineural invasion (PNI) in cutaneous squamous cell carcinoma (CSCC) has been associated with an increased risk of poor outcomes. Patients with PNI may present with clinical symptoms and/or radiologic evidence of PNI (clinical PNI [CPNI]), yet most patients are asymptomatic and PNI is often found on histologic examination (incidental PNI [IPNI]). Evidence-based estimates of the risks of disease-related outcomes comparing IPNI and CPNI are limited in the dermatology literature.

OBJECTIVES To review and synthesize outcomes data for patients with CSCC and CPNI or IPNI.

DATA SOURCES A systematic review was conducted in MEDLINE and EMBASE for English-language articles published since inception to November 11, 2016.

STUDY SELECTION All studies that reported a disease-related outcome (local recurrence, nodal metastasis, distant metastasis, or disease-specific death) of CSCCs with CPNI and IPNI were included.

DATA EXTRACTION AND SYNTHESIS Articles were screened for eligibility, and any possible discrepancies in this screening were resolved. Data extracted included study characteristics, tumor characteristics, treatments performed, and disease-related outcomes. Overall risks of disease-related outcomes were generated by pooling patients from eligible studies. χ^2 Statistics and Fisher exact tests were used to evaluate differences in disease-related outcomes.

MAIN OUTCOMES AND MEASURES Risks of disease-related outcomes and 5-year recurrence-free, disease-specific, and overall survival.

RESULTS A total of 12 studies containing 241 patients with CPNI and 381 patients with IPNI were included in the systematic review and analysis. The overall risks of local recurrence and disease-specific death were significantly higher in patients with CSCC and CPNI compared with those with CSCC and IPNI (local recurrence, 37% vs 17%; $P < .001$; disease-specific death, 27% vs 6%; $P < .001$). The risks of nodal metastasis and distant metastasis did not differ significantly by PNI classification. Patients with CSCC and CPNI had poorer mean 5-year recurrence-free survival and disease-specific survival compared with patients with IPNI (recurrence-free survival, 61% vs 76%; $P = .009$; disease-specific survival, 70% vs 88%; $P = .002$).

CONCLUSIONS AND RELEVANCE Patients with CSCC and CPNI are at an increased risk of local recurrence and disease-specific death compared with patients with CSCC and IPNI and have a 30% risk of death. Patients with PNI may benefit from increased long-term surveillance. Further studies are needed to establish standardized guidelines on follow-up and dermatologic surveillance in this high-risk patient population.

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Author Audio Interview

Supplemental content

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Cutaneous squamous cell carcinoma (CSCC) is the second most common malignant neoplasm in the US population, with an estimated 700 000 cases diagnosed annually.[1] Most CSCCs can be cured surgically. However, between 3.7% and 5.8% of CSCCs metastasize, most commonly to regional lymph nodes.[2-6] In addition, between 1.5% and 2.1% of CSCCs result in death.[2,4] It has been estimated that between 4000 and 9000 people in the United States die of CSCC annually.[7]

Perineural invasion (PNI) is a well-recognized risk factor in patients with CSCC, with reported incidence rates ranging from 2.5% to 14%.[8,9] Perineural invasion in patients with CSCC has been associated with poor prognosis, with local recurrence rates of 16% to 45% and nodal metastasis rates of 10% to 50%.[2,4,10,11] Risk factors for PNI in patients with CSCC include male sex, recurrent tumors, midface location, poorly differentiated histologic findings, and deep subclinical extension.[12,13]

A higher degree of nerve involvement is associated with poorer prognosis. For example, invasion of named nerves carries a poorer prognosis than does invasion of smaller unnamed nerves. Similarly, invasion of small-caliber nerves less than 0.1 mm in diameter does not portend a poorer prognosis unless other risk factors are present.[10,11] Accordingly, current National Comprehensive Cancer Network guidelines[14] recommend adjuvant radiotherapy for CSCCs with substantial PNI (involvement of more than a few small sensory nerve branches or large nerve involvement) regardless of surgical margin status. Adjuvant radiotherapy is also recommended in cases in which margins are positive after Mohs micrographic surgery or excision with complete circumferential peripheral and deep margin assessment. In patients with large or extensive perineural involvement, magnetic resonance imaging (MRI) is also recommended. However, detailed guidelines on follow-up schedules for such high-risk patients have not yet materialized.[14]

Most patients with CSCC and PNI present without clinical symptoms and no radiologic evidence of PNI. These tumors are classified as incidental or microscopic PNI (IPNI) because PNI is commonly identified on histologic examination of the CSCC at the time of surgery.[15-17] It is estimated that between 60% and 70% of patients with PNI present with incidental findings.[18,19] The role of adjuvant radiotherapy in patients with CSCC and IPNI is unclear, although some studies have reported 5-year local control rates of 77% to 87%.[17,20]

Some patients with CSCC and PNI present with clinical symptoms including pain, numbness, tingling, paralysis, or formication; these symptoms may or may not present with radiologic evidence of PNI.[19] Tumors with such symptoms or radiologically evident PNI are classified as clinical PNI (CPNI). It is estimated that between 30% and 40% of patients with PNI present with or develop clinical symptoms.[18] Patients with clinical symptoms of PNI and/or radiologic evidence of PNI extension often undergo aggressive surgical resection (if amenable), followed by adjuvant radiotherapy. However, evidence-based consensus regarding adjuvant treatments for patients with CSCC and CPNI is lacking.

Some studies in the head and neck literature have shown differences in disease-related outcomes between patients with CSCC and CPNI and patients with CSCC and IPNI.[20-22] However, to our knowledge, no such studies exist in the dermatology literature, even though most patients with CSCC are triaged to dermatologists and dermatologic surgeons. In this systematic review, we sought to examine current evidence regarding the risk of poor outcomes (local recurrence, nodal and distant metastasis, death from disease, and overall death) in patients with CSCC and CPNI or IPNI and investigate whether significant differences exist in the risk of these outcomes between patients with CPNI and patients with IPNI. Differences in 5-year recurrence-free survival (RFS), disease-specific survival (DSS), and overall survival (OS) were also examined.

Key Points

Question Is there a risk difference in disease-related outcomes between patients with cutaneous squamous cell carcinoma and clinical perineural invasion vs patients with cutaneous squamous cell carcinoma and incidental perineural invasion?

Finding In this systematic review and pooled analysis, outcomes data from the current literature were compared among patients with cutaneous squamous cell carcinoma and clinical perineural invasion or incidental perineural invasion. Patients with cutaneous squamous cell carcinoma and clinical perineural invasion had a significantly higher risk of local recurrence and death from disease than did patients with cutaneous squamous cell carcinoma and incidental perineural invasion.

Meaning Patients with cutaneous squamous cell carcinoma and clinical perineural invasion are at a greater risk of poor outcomes compared with those with cutaneous squamous cell carcinoma and incidental perineural invasion and may benefit from enhanced long-term surveillance.

Methods

Search Strategy

A systematic review was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.[23] The MEDLINE and EMBASE databases were searched on November 11, 2016 using the search terms *squamous cell carcinoma perineural nerve invasion*, *squamous cell carcinoma clinical perineural nerve invasion*, and *squamous cell carcinoma incidental perineural nerve invasion*. Search results were restricted to English-language studies only. No publication date or study design restrictions were used.

Selection Criteria

Eligible studies included those that reported disease-related outcomes (local recurrence, nodal metastasis, distant metastasis, or death from disease) of patients with CSCC and CPNI or IPNI. Studies reporting outcomes of CSCC, basal cell carcinoma, and basosquamous carcinoma were included only if outcomes by tumor subtype were reported. For such studies, only cases of CSCC and subsequent outcomes were included in the analysis. Case reports were included only if they reported more than 5 patients. Studies were excluded if they did not report the type of PNI (clinical vs incidental), if they did not report

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any disease-related outcomes, if they reported noncutaneous PNI or PNI of unknown tumor type, and if they did not report any original data (review articles and editorials).

Study Selection and Data Extraction

Articles obtained from the search were independently screened for eligibility by 2 authors (P.S.K. and E.S.R.). Discrepancies were resolved by a third author (C.D.S.). Reference lists of all eligible articles were reviewed to locate additional studies that may have been missed during the preliminary search. Each reviewer also assessed the quality of reporting and the risk of bias for included articles. The following data were extracted from each article: characteristics of study design (including number of patients, number of primary and recurrent tumors, anatomical restrictions, median follow-up time, and time frame of data collection), characteristics of tumors (including tumor histologic characteristics, type of PNI, and grading of PNI based on nerve and zonal involvement), treatments performed (including surgical margin status, imaging modalities, primary treatments, adjuvant treatments, and treatment complications), and disease-related outcomes (including local recurrence, nodal metastasis, distant metastasis, death from CSCC, and overall death). Survival statistics were also recorded, including RFS (survival free from any local, nodal, or distant recurrence), DSS (survival free from CSCC-related death), and OS (survival free from any cause of death).

Statistical Analysis

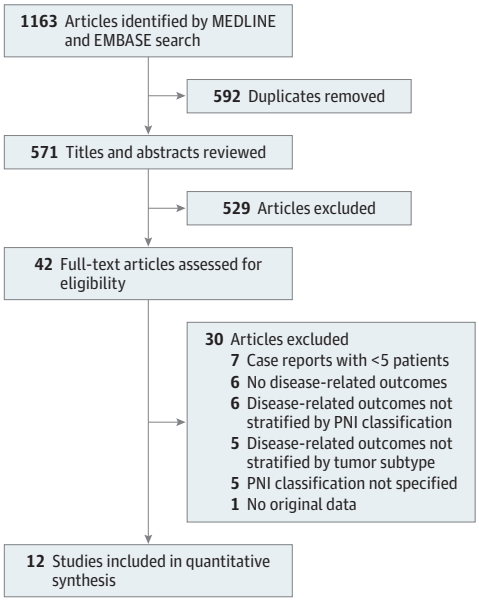
Overall risks of disease-related outcomes were generated by pooling patients from eligible studies. Reported survival statistics from included studies were averaged to produce overall estimates. The χ^2 and Fisher exact tests were used to evaluate differences in disease-related outcomes and survival between patients with CPNI and patients with IPNI. All statistical analyses were conducted using Stata, version 12.0 (StataCorp).

Results

The **Figure** shows the study selection process. The database search yielded 1163 articles. After duplicates were removed, 571 titles and abstracts were reviewed, and 529 were subsequently excluded because they did not relate to the study topic or did not contain the required information. The remaining 42 articles were reviewed in detail for eligibility. After exclusion criteria were applied, 30 articles were excluded, leaving 12 studies included in the systematic review.[10,21,24-33]

The 12 studies included were published between 1995 and 2016, with data collection between 1965 and 2013. Most of the studies (8 [67%]) were retrospective, and the remainder were case series or case reports (4 [33%]) (**Table 1**).[10,21,24-34] The median follow-up time ranged from 19 to 89 months. The studies included a total of 640 tumors with CPNI or IPNI in 622 patients. Most CSCC tumors were located primarily in the head and neck region (8 studies [67%]). A total of 133 cases of CSSC (20.8%) were recurrent at the time of data collection.

Figure. Flowchart of Studies Included in the Systematic Review



PNI indicates perineural invasion.

Clinical PNI

A total of 6 studies with 241 patients reported outcomes of CPNI cases.[21,24,26,27,30,33] The risk of local recurrence ranged from 19% (7 of 36 patients) to 73% (8 of 11 patients), while the risk of nodal metastasis ranged from 2% (1 of 50 patients) to 10% (2 of 21 patients), with 1 study reporting no such events (**Table 2**).[10,21,24-33] Distant metastasis was rare, with 1 study reporting a 2% risk (1 of 50 patients) and 4 studies reporting no such events. Among studies that reported death from disease, the risk ranged from 19% (4 of 21 patients) to 42% (15 of 36 patients). The overall risk of local recurrence was 37% (89 of 241 patients), the overall risk of nodal metastasis was 6% (12 of 206 patients), the overall risk of distant metastasis was 0.5% (1 of 206 patients), and the overall risk of death from disease was 27% (29 of 107 patients). In terms of survival statistics, the 5-year RFS ranged from 59% to 62%, the 5-year DSS ranged from 64% to 75%, and the 5-year OS ranged from 64% to 68%. The mean overall 5-year RFS was 61%, the mean overall 5-year DSS was 70%, and the mean overall 5-year OS was 66%.

Most of the 206 cases of CSCC with CPNI were treated with surgery and postoperative radiotherapy (141 [68.4%]). A total of 34 cases (16.5%) were treated with radiotherapy alone. Less common treatment modalities included radiotherapy and chemotherapy (11 [5.3%]); other combination therapy (7 [3.4%]); surgery and preoperative radiotherapy (5 [2.4%]); surgery, postoperative radiotherapy, and chemotherapy (4 [1.9%]); and surgery alone (4 [1.9%]). Of the 115 studies that reported surgical margin status, 58 cases (50.4%) had clear margins, 44 (38.3%) had positive margins, and 13 (11.3%) had close margins (<5 mm).

Information on patterns of perineural spread was reported for 246 CPNI cases. Most of the cases had trigeminal nerve (V1 [ophthalmic], V2 [infraorbital], or V3 [mandibular]) involvement (171 [69.5%]), while multiple trigeminal nerves

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Table 1. Summary of Studies Included in the Systematic Review

Source	Study Design	Quality of Evidence[a]	Period of Data Collection	Follow-up, Median (Range), mo	Anatomical Restrictions	Cases, No.	Recurrent at Presentation, No.
Sapir et al,[33] 2016	RC	3	2000-2013	23 (1-127)	Head and neck	35	NR
Warren et al,[24] 2016	RC	3	2000-2011	50 (9-151)	Head and neck	50	29
Kropp et al,[25] 2013	RC	3	1987-2009	50 (2-258)	Head and neck	32	8
Carter et al,[10] 2013	RC	3	1998-2008	47 (2-120)	None	113	0
Panizza et al,[26] 2012	RC	3	1996-2006	NR	Head and neck	21	NR
Balamucki et al,[21] 2012	RC	3	1965-2007	48 (2-312)	Head and neck	185	71
Solares et al,[27] 2012	CS	4	1996-2006	19 (NR)	Head and neck	36	NR
Lin et al,[28] 2012	RC	3	1991-2004	52 (3-201)	None	133	24
DeAmbrosis et al,[29] 2010	CS	4	NR	89 (25-218)	None	10	0
Gluck et al,[30] 2009	RC	3	2000-2007	NR	Head and neck	11	NR
Geist et al,[31] 2008	CS	4	2001-2006	NR	None	8	1
Ampil et al,[32] 1995	CR	5	1965-1991	45 (18-201)	Head and neck	6	0

Abbreviations: CR, case report; CS, case series; NR, not recorded; RC, retrospective cohort.

[a] Assessed using the Quality Rating Scheme for Studies and Other Evidence,[34] where 1 indicates properly powered and conducted randomized clinical trial or systematic review with meta-analysis; 2, well-designed controlled trial without randomization or prospective comparative cohort trial; 3, case-control studies or retrospective cohort study; 4, case series with or without intervention or cross-sectional study; and 5, opinion of respected authorities or case reports.

were involved in only 3 cases (1.2%). The facial nerve (VII) was involved in 70 cases (28.5%), while the vestibulocochlear nerve (VIII) was involved in only 2 cases (0.8%). The most common symptoms of CPNI included sensory deficits, cranial nerve deficits, pain, motor deficits, and formication. Only 4 studies reported details of the use of radiologic imaging. Magnetic resonance imaging was the main modality used, with 71 of 74 imaged cases (95.9%) having positive findings. A computed tomographic scan was used to determine the extent of PNI in patients for whom MRI was contraindicated.

Incidental PNI

A total of 7 studies with 381 patients reported outcomes in IPNI cases.[10,21,25,28,29,31,32] The risk of local recurrence ranged from 13% (4 of 32 patients) to 23% (22 of 97 patients), while the risk of nodal metastasis ranged from 9% (15 of 165 patients) to 17% (1 of 6 patients) (Table 2).[10,21,24-33] One study had no occurrences of local recurrence or nodal metastasis. The risk of distant metastasis ranged from 1% (3 of 228 patients) to 10% (1 of 10 patients), with 2 studies reporting no events. Among studies that reported death from disease, the risk ranged from 0% to 7% (7 of 95 patients). The overall risk of local recurrence was 17% (64 of 381 patients), the overall risk of nodal metastasis was 10% (40 of 381 patients), the overall risk of distant metastasis was 3% (10 of 381 patients), and the overall risk of death from disease was 6% (7 of 111 patients). In terms of survival statistics, the 5-year RFS ranged from 72% to 78%, 5-year DSS ranged from 84% to 91%, and 1 study reported a 5-year OS of 43%. The mean overall 5-year RFS was 76%, the mean overall 5-year DSS was 88%, and the mean overall 5-year OS was 43%.

Of the studies that reported information on IPNI nerve caliber, 40% reported large (≥0.1 mm) or extensive PNI.[10,28,29] Most of the 399 IPNI cases were treated with surgery and postoperative radiotherapy (233 [58.4%]). Less common treatment modalities included surgery only (83 [20.8%]); Mohs surgery and radiotherapy (42 [10.5%]); Mohs surgery only (15 [3.8%]); radiotherapy only (10 [2.5%]); surgery, postoperative radiotherapy, and chemotherapy (6 [1.5%]); other combination therapy (6 [1.5%]); and surgery and preoperative radiotherapy (4 [1.0%]).

Comparison of Disease-Related Outcomes

The eTable in the Supplement summarizes outcomes of patients by PNI classification. There was no significant difference in the risk of nodal and distant metastasis by PNI classification. However, patients with CPNI had higher overall risks of local recurrence and death from CSCC compared with patients with IPNI (local recurrence, 37% vs 17%; $P < .001$; death from CSCC, 27% vs 6%; $P < .001$). In addition, patients with IPNI had a higher 5-year DSS rate compared with patients with CPNI (RFS, 76% vs 61%; $P = .009$; DSS, 88% vs 70%; $P = .002$). Conversely, 5-year OS was higher for patients with CPNI compared with patients with IPNI (66% vs 43%; $P = .003$). One study compared outcomes of IPNI and CPNI cases.[21] Exclusion of this study from the overall analysis did not significantly alter the results summarized.

Quality of Evidence for Included Studies

Table 3[10,21,24-34] provides an assessment of the quality of evidence for the included studies. Most studies were nonrandomized, observational cohort studies and were thus graded as 2A (weak recommendation; limited-quality, patient-oriented evidence) or 2B (weak recommendation; low-quality evidence).[34] Although limited by the lack of randomized clinical trials, the data derived from these studies were consistent in showing a decreased risk for disease-related outcomes among patients with CSCC and IPNI vs those with CSCC and CPNI. Based on the consistency of these results, adjuvant radiotherapy and regular posttreatment surveillance may be indicated for all patients with CPNI regardless of PNI caliber and for patients with IPNI with large-caliber PNI (≥0.1 mm). Patients with IPNI with small-caliber PNI (<0.1 mm) may not require adjuvant radiotherapy or regular posttreatment surveillance because previous studies have shown that this group

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Table 2. Summary of Cutaneous Squamous Cell Carcinoma Outcomes by Perineural Invasion Classification

Source	Patients, No.	Risk of Poor Outcomes, No. (%)				5-y Survival Statistics, %			Treatment, No. (%)
		LR	NM	DM	DSD	RFS	DSS	OS	
Clinical perineural invasion									
Warren et al,[24] 2016	50	16 (32)	1 (2)	1 (2)	10 (20)	62	75	64	A (50 [100])
Panizza et al,[26] 2012	21	5 (24)	2 (10)	0	4 (19)	59	64	68	A (18 [86]), B (3 [14])
Solares et al,[27] 2012	36	7 (19)	3 (8)	0	15 (42)	NR	NR	NR	A (20 [56]), C (5 [14]), D (4 [11]), I (7 [19])
Balamucki et al,[21] 2012	88	40 (45)	6 (7)	0	NR	NR	NR	NR	A (42 [48]), B (2 [2]), C (29 [33]), E (11 [12]), F (4 [5])
Gluck et al,[30] 2009	11	8 (73)	0	0	NR	NR	NR	NR	A (11 [100])
Sapir et al,[33] 2016	35	13 (37)	NR	NR	NR	NR	NR	NR	NR
Total	241	89 (37)	12/206 (6)	1/206 (0.5)	29/107 (27)	61	70	66	A (141/206 [68]), B (5/206 [2]), C (34/206 [19]), E (11/206 [6]), F (4/206 [2]), D (4/206 [2]), I (7/206 [3])
Incidental perineural invasion									
Carter et al,[10] 2013	95	18 (19)	11 (12)	1 (1)	7 (7)	72	91	43	A (15 [13]), C (1 [1]), D (74 [65]), G (2 [2]), H (15 [13]), I (6 [6])
Kropp et al,[25] 2013	32	4 (13)	3 (9)	2 (6)	NR	77	84	NR	G (32 [100])
Lin et al,[28] 2012	133	17 (13)	12 (9)	2 (1)	NR	78	NR	NR	A (128 [96]), C (5 [4])
Balamucki et al,[21] 2012	97	22 (23)	12 (12)	4 (4)	NR	NR	NR	NR	A (97 [85]), B (4 [4]), C (4 [4]), F (7 [7])
DeAmbrosis et al,[29] 2010	10	2 (20)	1 (10)	1 (10)	0	NR	NR	NR	A (6 [60]), D (4 [40])
Geist et al,[31] 2008	8	0	0	0	NR	NR	NR	NR	G (8 [100])
Ampil et al,[32] 1995	6	1 (17)	1 (17)	0	0	NR	NR	NR	A (1 [17]), D (5 [83])
Total	381	64 (17)	40 (10)	10 (3)	7/111 (6)	76	88	43	A (232/399 [60]), B (4/399 [1]), C (10/399 [3]), D (83/399 [20]), E (0), F (7/399 [2]), G (42/399 [12]), H (15/399 [4]), I (6/399 [1])[a]

Abbreviations: A, surgery and postoperative radiotherapy; B, surgery and preoperative radiotherapy; C, radiotherapy only; D, surgery only; DM, distant metastasis; DSD, disease-specific death; DSS, disease-specific survival; E, radiotherapy and chemotherapy; F, surgery and postoperative radiotherapy and chemotherapy; G, Mohs micrographic surgery and radiotherapy; H, Mohs micrographic surgery only; I, other combination therapy; LR, local recurrence; NM, nodal metastasis; NR, not recorded; OS, overall survival; RFS, recurrence-free survival.

[a] These numbers total 399 because Carter et al had patients with multiple incidental perineural invasion tumors (113 tumors in 95 patients).

of patients has a low risk of poor outcomes in the absence of other risk factors.[10,35]

Discussion

In this systematic review of 12 studies and 640 CSCC tumors with PNI, patients with CPNI were significantly more likely to develop local recurrence and die of CSCC compared with patients with IPNI, with patients with CPNI having a 30% risk of death. No difference in the risk of nodal or distant metastasis was observed between patients with CPNI and those with IPNI. In addition, patients with CPNI had poorer mean 5-year RFS and DSS compared with patients with IPNI. Although mean 5-year OS was better among patients with CPNI than among those with IPNI, these mean values were acquired from study cohorts that likely contained older patient populations, which are at an increased risk of mortality from multiple causes, not just CSCC.

These findings are consistent with those of previous studies comparing CPNI and IPNI outcomes. However, to our knowledge, this is the first study to summarize all available outcomes data on CSCC by PNI classification and the first to report such a comparison in the dermatology literature. Prior studies have suggested that the presence of clinical symptoms of PNI is independently associated with local control rates in head and neck nonmelanoma skin cancers (CSCC, basal cell carcinoma, and basosquamous carcinoma).[20-22] A study of 118 patients with CSCC and basal cell carcinoma found significant differences in DSS and OS between IPNI and CPNI cases (DSS, 90% vs 76%; $P = .002$; OS, 69% vs 57%; $P = .02$).[22] Another study of 135 patients with nonmelanoma skin cancer treated with radiotherapy alone or combined with surgery found significantly poorer 5-year local control rates among patients with CPNI vs those with IPNI (55% vs 87%; $P = .006$).[20] These studies were not included in our analysis because they did not separate disease-related outcomes by tumor subtype. One study[21] that was included

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Table 3. Summary of Recommendations and Quality of Evidence for Included Studies

Source	Grade of Recommendation[a]	Quality of Evidence[b]	Summary of Recommendations
Clinical PNI			
Sapir et al,[33] 2016	2A	B	Adjuvant radiotherapy and posttreatment surveillance recommended after considering patient preference, age, and comorbidities
Balamucki et al,[21] 2012	2A	B	
Warren et al,[24] 2016	2A	B	
Panizza et al,[26] 2012	2A	B	
Gluck et al,[30] 2009	2A	B	
Solares et al,[27] 2012	2B	C	
Incidential PNI			
Kropp et al,[25] 2013	2A	B	Small-caliber PNI (<0.1 mm): adjuvant radiotherapy and posttreatment surveillance not recommended; large-caliber PNI (≥0.1 mm): adjuvant radiotherapy and posttreatment surveillance recommended after considering patient preference, age, and comorbidities
Carter et al,[10] 2013	2A	B	
Balamucki et al,[21] 2012	2A	B	
Lin et al,[28] 2012	2A	B	
DeAmbrosis et al,[29] 2010	2B	C	
Geist et al,[31] 2008	2B	C	
Ampil et al,[32] 1995	2B	C	

Abbreviation: PNI, perineural invasion.

[a] Based on Robinson et al,[34] in which 1 indicates strong recommendation and high-quality, patient-oriented evidence; 2A, weak recommendation and limited-quality, patient-oriented evidence; and 2B, weak recommendation and low-quality evidence.

[b] A indicates systematic review or meta-analysis of good-quality cohort studies that can apply to most patients; B, systematic review or meta-analysis of lower-quality cohort studies with inconsistent results that may vary depending on circumstances, patients, or societal values; retrospective cohort studies; case-control studies; and C, consensus guidelines, usual practice, expert opinion, and case series.

in our review analyzed 216 patients with CSCC and basal cell carcinoma and CPNI and IPNI. Those authors[21] reported disease-related outcomes by tumor subtype but not survival statistics (overall 5-year DSS of 73% and overall 5-year OS of 55% for IPNI and overall 5-year DSS of 64% and overall 5-year OS of 54% for CPNI). Exclusion of this study's disease-related outcomes from the systematic review did not significantly change the results, indicating that it is not driving the results herein.

The results of this study underscore the importance of a thorough evaluation for signs and symptoms of PNI during the pretreatment clinical examination. Patients with CPNI may warrant aggressive treatment and close surveillance for recurrence following treatment. The elevated risks of both local recurrence in and disease-specific death of patients with CPNI are consistent with reports that most CSCC-related deaths occur owing to uncontrolled locoregional recurrence rather than distant metastasis.[18,36-38]

Based on the evidence summarized herein, we believe that informed decision making regarding adjuvant therapy and posttreatment surveillance among patients with CPNI is justified. Continued long-term surveillance via MRI every 6 months for 2 to 3 years after treatment should be considered after taking patient preference, age, and comorbidities into account. Early detection and retreatment has the potential to increase survival, although the utility of radiologic surveillance has not been well quantified. A study of 12 patients with CPNI, 11 of whom were evaluated with MRI before therapy, found that the results of MRI established the diagnosis of CPNI for 6 patients, showed the progression of disease in 3 asymptomatic patients, and revealed the full extent of disease in 6 patients.[30] A recent study analyzing a cohort of patients with high-stage CSCC, of which 37% had PNI, found that patients who underwent imaging were less likely to experience disease-related outcomes compared with patients who did not undergo imaging.[39]

The benefits of radiotherapy for patients with PNI have been well established.[14,40] Current National Comprehensive Cancer Network guidelines recommend adjuvant radiotherapy for high-risk CSCC, which includes CSCC with substantial PNI (large-caliber PNI and involvement of more than just a few small sensory nerve branches).[14] Studies on adjuvant chemotherapy are lacking. To our knowledge, only 1 study has evaluated the association between adjuvant chemotherapy and high-risk head and neck CSCC.[41] This retrospective study of 61 patients, 30 of whom had PNI, found that those who received adjuvant chemotherapy had longer median RFS compared with those who received adjuvant radiotherapy (40.3 vs 15.4 months; $P = .01$). On multivariable analysis, treatment with adjuvant chemotherapy was associated with a 69% reduced risk of recurrence (hazard ratio, 0.31; 95% CI, 0.13-0.78; $P = .01$), after adjusting for tumor differentiation, PNI, and tumor and nodal classification. There was no difference in OS between the 2 treatment groups. These results are consistent with those of 2 large randomized clinical trials of patients with mucosal squamous cell carcinoma of the head and neck.[42,43] A phase 2 trial is currently ongoing for a fully human monoclonal antibody to programmed death-1 in patients with advanced CSCC. If immunotherapy is shown to be efficacious for advanced CSCC, future studies could be conducted to examine its role in an adjuvant setting.

Even though patients with CPNI are more likely to have poor outcomes, patients with IPNI are also at risk for poor outcomes, and in the appropriate patient, after taking patient preference, age, and comorbidities into consideration, adjuvant therapy may be indicated to minimize the risk of local recurrence and metastasis. Adjuvant therapy may particularly benefit patients with IPNI who have large-caliber PNI (≥0.1 mm) because previous studies have shown that patients with IPNI with small-caliber PNI (<0.1 mm) have a low risk of poor outcomes in the absence of other prognostic risk factors.[10,35]

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Limitations

This study is limited in that it included only retrospective studies and case series. To our knowledge, no prospective cohort studies exist that examine differences in disease-related outcomes in patients with CSCC with PNI. The included studies therefore represent the best evidence available on the prognosis of patients with CSCC with IPNI or CPNI. Another potential limitation is that the data-collection period for included studies ranged from 1965 to 2013, during which time the definition of PNI and the detection and treatment modalities may have changed. Nevertheless, it is unlikely that the exclusion of older studies would have significantly altered the results of the analysis. Finally, multiple included studies were from the same institutions. However, duplicate reports of the same or overlapping patient cohorts were excluded.

Conclusions

Although PNI is relatively uncommon in patients with CSCC, it is associated with an increased risk of recurrence and death. The data herein demonstrate that patients with CPNI (ie, those with radiographic evidence of PNI and/or clinical symptoms) have a significantly higher risk of local recurrence and death from CSCC compared with patients with IPNI (ie, asymptomatic PNI, detected on histopathologic examination). In addition, patients with CPNI have poorer 5-year RFS and DSS compared with those with IPNI. Although there are currently no guidelines on monitoring these patients, patients with PNI, particularly CPNI, may benefit from posttreatment surveillance. Further studies are needed to establish evidence-based guidelines for optimal treatment of various degrees of PNI and radiologic follow-up.

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Study concept and design: Karia, Ruiz, Schmults.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: Karia, Schmults.
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